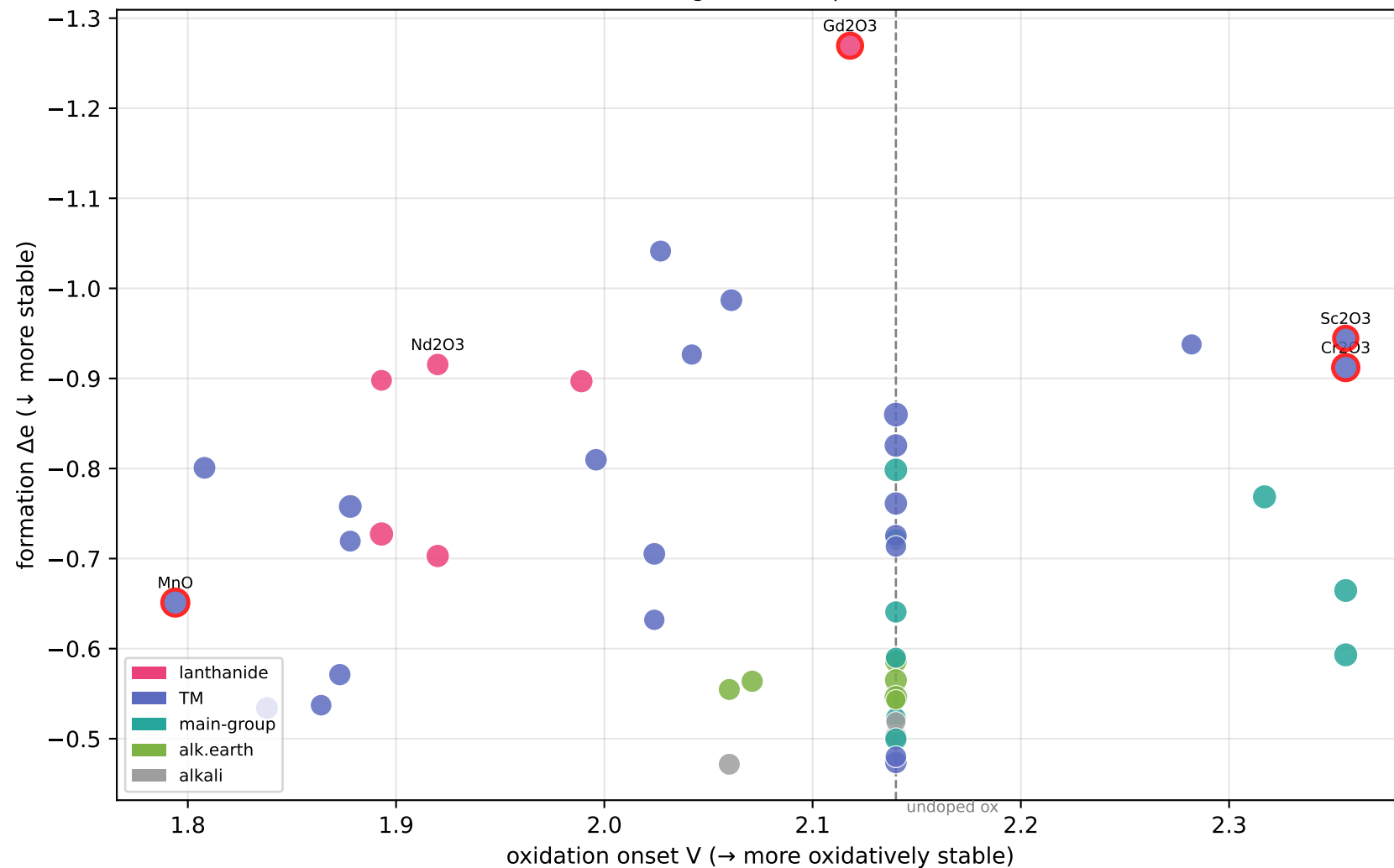
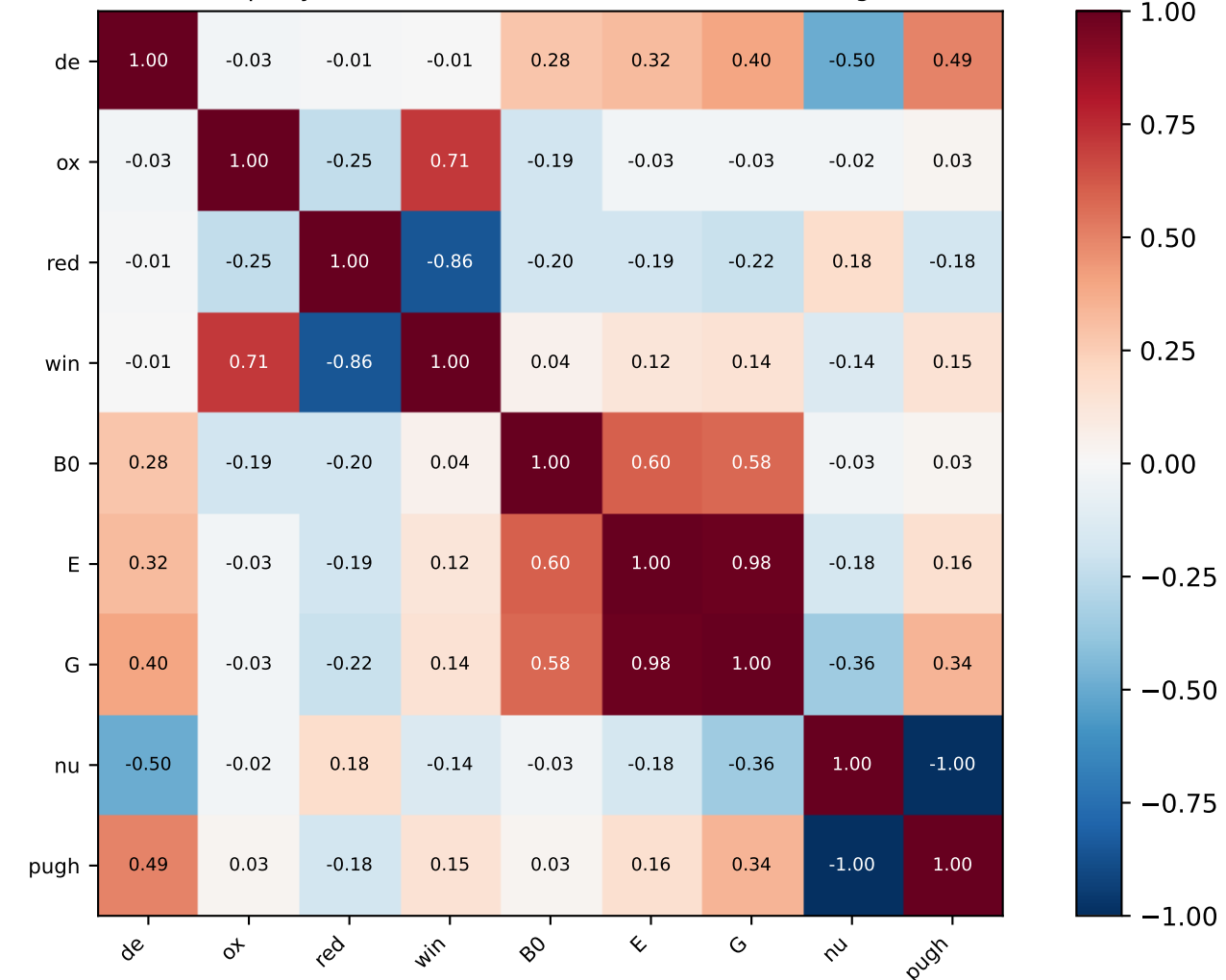


Cascade v23 — multi-dimensional synthesis (stability × mechanics × oxidation window), 47 dopants

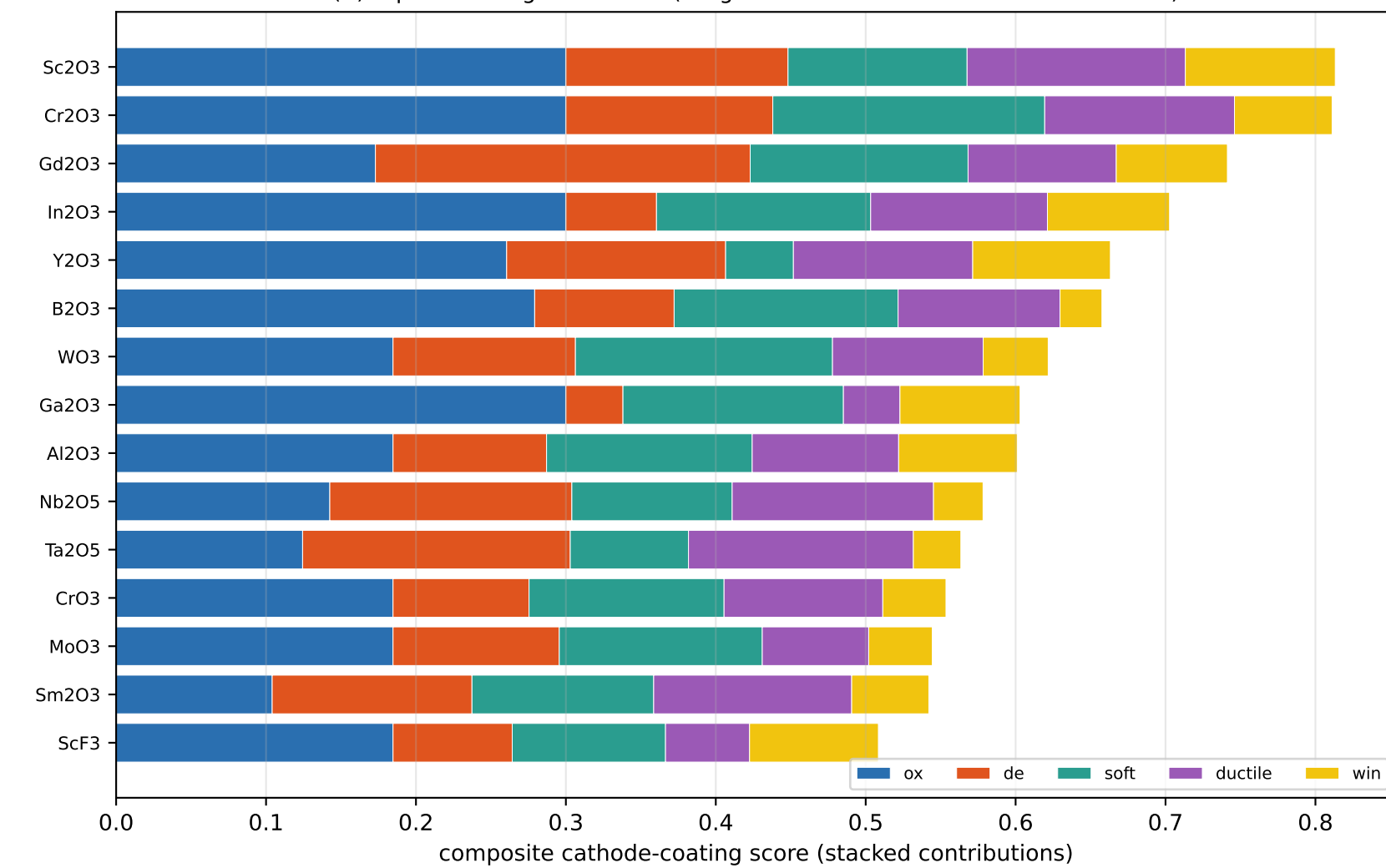
(A) Cathode-coating map — ideal = right(ox-stable)+down(stable)+big(soft)
red ring = Pareto-optimal



(B) Property correlations (Pearson r) — what moves together



(C) Top-15 coating candidates (weights: ox.30 de.25 soft.20 duct.15 win.10)



(D) Anode vs cathode stability — late-TM (top-right cluster) bad on both

